Biomedical Signal Processing

ECG Signal Decomposition Using Fourier Analysis

By: SHARIF AHMADIFAR

Supervisors: Roberto Sassi, Massimo Walter Rivolta

July, 2025

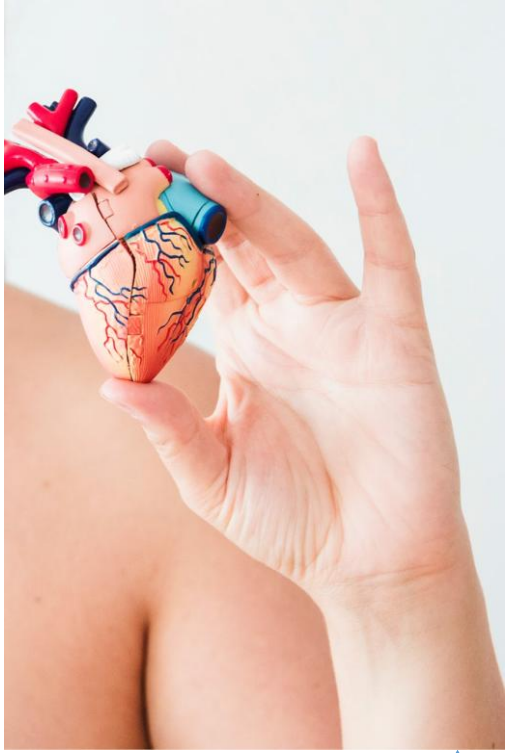
Table of contents

01 Introduction

02 Fourier Series Expansion

03 ECG Example

04 Conclusion



01

Introduction

Reference Paper: ECG Signal Decomposition Using Fourier Analysis

Kheirati Roonizi and Sassi
EURASIP Journal on Advances in Signal Processing (2024) 2024:79
https://doi.org/10.1186/s13634-024-01171-x

EURASIP Journal on Advances in Signal Processing

RESEARCH Open Access

ECG signal decomposition using Fourier analysis

Arman Kheirati Roonizi[1,2*] and Roberto Sassi[2]

*Correspondence: armankheirati@unimi.it; a-kheirati@fasa.ac.ir

[1] Department of Computer Science, Faculty of Science, Fasa University, Fasa 74616-86131, Iran
[2] Department of Computer Science, University of Milan, Via Celoria 18, Milan, Italy

Abstract

This paper explores the Fourier decomposition method to approximate the decomposition of electrocardiogram (ECG) signals into their component waveforms, such as the QRS-complex and T-wave. We compute expansion coefficients using the ℓ_1 Fourier transform and the traditional ℓ_2 Fourier transform. Numerical examples are presented, and the analysis focuses on ECG signals as a real-world application, comparing the performance of the ℓ_1 and ℓ_2 Fourier transforms. Our results demonstrate that the ℓ_1 Fourier transform significantly enhances the separation of ECG signal components, such as the QRS-complex and T-wave. This improvement is attributed to a notable reduction in the Gibbs phenomenon introduced by the Fourier-series expansion when using the ℓ_1 Fourier transform, as opposed to the traditional ℓ_2 Fourier transform.

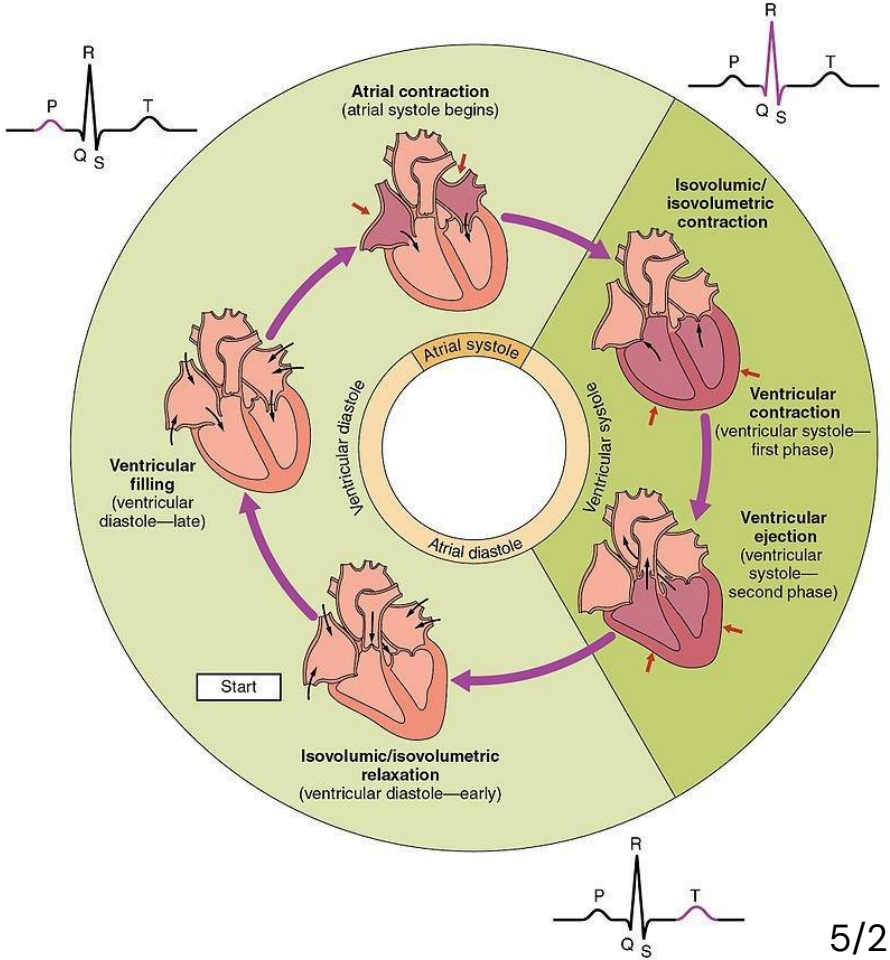
Keywords: ℓ_1 Fourier analysis, ℓ_1 Fourier analysis, Signal decomposition, ECG components

All simulations, comparisons, and interpretations in this presentation are based on the methodology and findings of this published article.

Why ECG Analysis Matters?

- ✓ ECG signals contain multiple overlapping waveforms: P, QRS, and T waves
- ✓ These components reflect different cardiac activities such as **depolarization** and **repolarization**
- ✓ The frequency content of T-wave and QRS-complex overlaps, making it difficult to separate them using classical tools.

- ✓ Accurate decomposition helps improve diagnosis and analysis in cardiology.

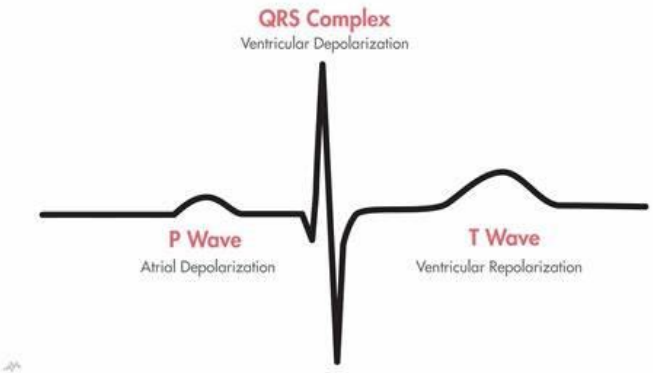


Problem with Traditional Fourier Transform ℓ_2

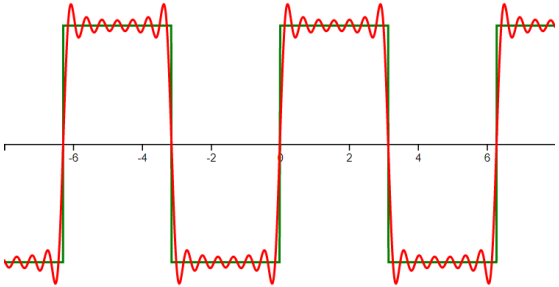
ℓ_2 –based Fourier analysis minimizes the squared error.

Squared error is sensitive to outliers and impulsive signals.

like QRS peaks



- This results in Gibbs phenomenon and leakage of sharp components into slow ones.



- ✓ ℓ_2 norm suffers from Gibbs phenomenon,
- ✓ sharp transitions leak into low-frequency components

A Better Solution: ℓ_1 Fourier Transform

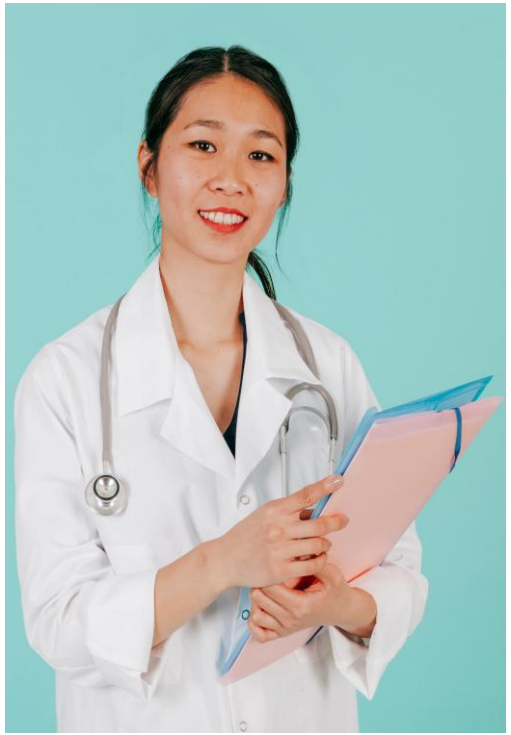
- ✓ minimizes absolute error
- ✓ less sensitive to outliers

- ✓ Better for signals with impulsive
- ✓ Reduces Gibbs phenomenon

ℓ_1
Fourier transform

- ✓ Efficiently estimates slow component (T-wave)
- ✓ suppresses QRS

- ✓ improves component separation
- ✓ Reduces Gibbs phenomenon



02

Furier Series Expansion

[redacted]

Fourier-series expansion-based signal representation

$$x_M[n] = \sum_{k=0}^{M-1} c_k \phi_k[n]$$

$$n = \{0,\ 1, \cdots,\ N-1\}$$

$$\phi_k[n] = \exp(i\tfrac{2\pi k}{N} n)$$

$$\boldsymbol{e} = \boldsymbol{x} - \boldsymbol{x}_M$$

$$\boldsymbol{x}_M = \boldsymbol{\Phi}\boldsymbol{c}$$

- ✓ where $\boldsymbol{\Phi}$ is an N × M matrix,
- ✓ $\boldsymbol{x}_M = (x_M[0], \cdots,\ x_M[N])^T$ is a N-length vector,
- ✓ $\boldsymbol{c} = (c_0, \cdots,\ c_{M-1})^T$ is a M-length vector

ℓp Fourier transform:

$$\underset{\boldsymbol{c}}{\operatorname{argmin}} \|\boldsymbol{x} - \boldsymbol{x}_M\|_p$$

ℓ2 Fourier transform: $\underset{\boldsymbol{c}}{\operatorname{argmin}} \|\boldsymbol{x} - \boldsymbol{\Phi}\boldsymbol{c}\|_2$

ℓ1 Fourier transform: $\underset{\boldsymbol{c}}{\operatorname{argmin}} \|\boldsymbol{x} - \boldsymbol{\Phi}\boldsymbol{c}\|_1$

Fourier-series expansion-based signal representation

ℓ_2 Fourier transform: $\underset{\boldsymbol{c}}{\operatorname{argmin}} \|\boldsymbol{x} - \boldsymbol{\Phi}\boldsymbol{c}\|_2$

$$\boldsymbol{c} = (\boldsymbol{\Phi}^H \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^H \boldsymbol{x}$$

- ✓ Where $\boldsymbol{\Phi}^H$ is the complex conjugate transpose (Hermitian transpose) of $\boldsymbol{\Phi}$
- ✓ $\boldsymbol{\Phi}^H \boldsymbol{\Phi}$ ▶ identity matrix ▶ orthogonal Fourier basis functions
- ✓ Simplified ▶ $\boldsymbol{c} = \boldsymbol{\Phi}^H \boldsymbol{x}$

Fourier-series expansion-based signal representation

ℓ_1 Fourier transform: $\underset{\boldsymbol{c}}{\operatorname{argmin}} \|\boldsymbol{x} - \boldsymbol{\Phi}\boldsymbol{c}\|_1$

$$\boldsymbol{c}^{(r+1)} = \underset{\boldsymbol{c}^{(r)}}{\operatorname{argmin}} \left(\boldsymbol{x} - \boldsymbol{x}_M^{(r)}\right)^T \boldsymbol{\Lambda}_r^{-1} \left(\boldsymbol{x} - \boldsymbol{x}_M^{(r)}\right) + \frac{\gamma_r}{2}$$

- ✓ iterative numerical algorithm ▶ majorization minimization (MM)
- ✓ Where $\boldsymbol{\Lambda}_r = \begin{pmatrix} |\boldsymbol{e}_r[0]| & 0 & \dots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & |\boldsymbol{e}_r[N]| \end{pmatrix}$, $\gamma_r = \|\boldsymbol{x} - \boldsymbol{x}_M^{(r)}\|_1$, $\boldsymbol{x}_M^{(r)} = \boldsymbol{\Phi}\boldsymbol{c}^{(r)}$, $\boldsymbol{e}_r = \boldsymbol{x} - \boldsymbol{x}_M^{(r)}$

$$\boldsymbol{c}^{(r+1)} = \left(\boldsymbol{\Phi}^H \boldsymbol{\Lambda}_r^{-1} \boldsymbol{\Phi}\right)^{-1} \boldsymbol{\Phi}^H \boldsymbol{\Lambda}_r^{-1} \boldsymbol{x}$$

Fourier-series expansion-based signal representation

$$x_M[n] = \sum_{k=0}^{M-1} c_k \phi_k[n], \quad M < N$$

▶ truncated Fourier-series expansion ▶ low-pass filter

Fourier-series decomposition methods ▶ decomposition of a signal

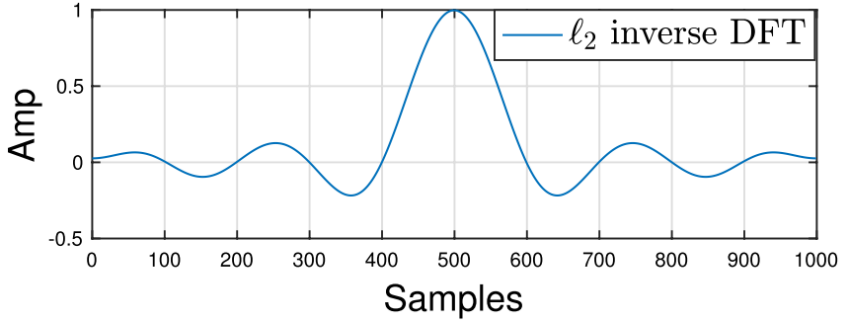
employ ℓ_1 and ℓ_2 Fourier transform ▶ compute the expansion coefficients

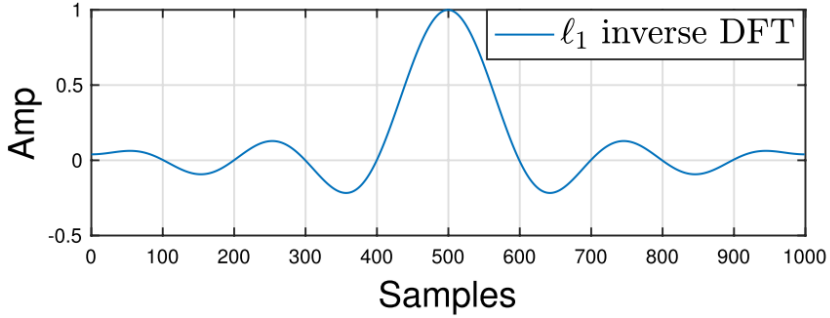
Numerical example

ECG example

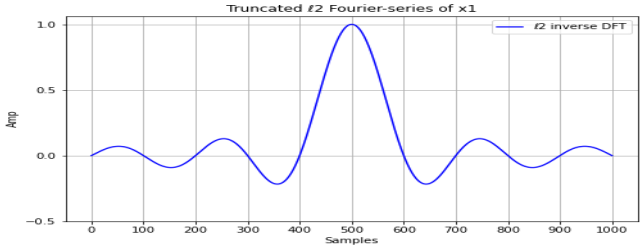
Fourier-series expansion-based signal representation

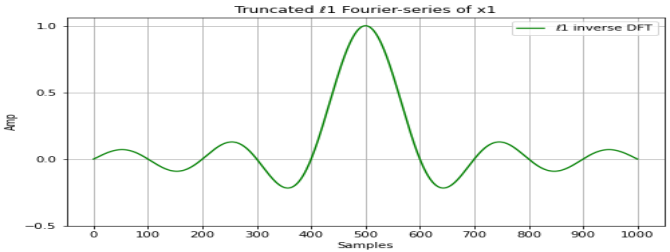
$x_1[n] = sinc[n], n = 0, \cdots, 1000$





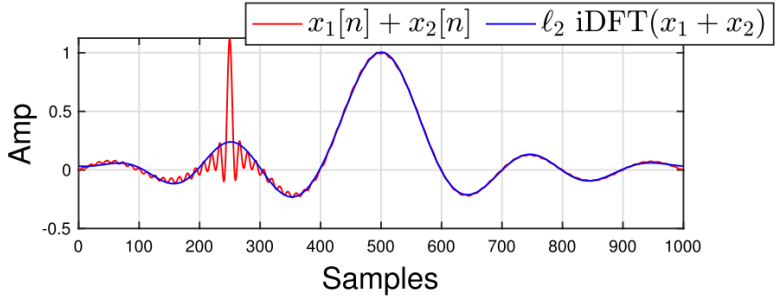
My simulation results

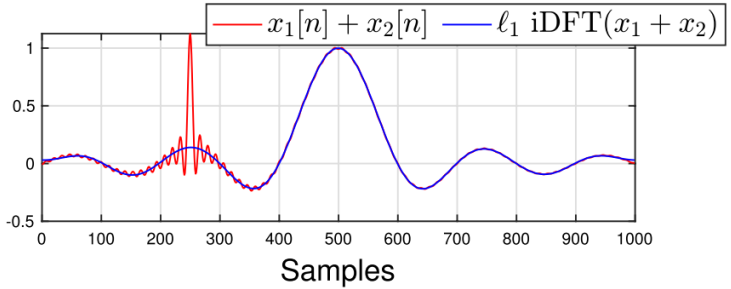




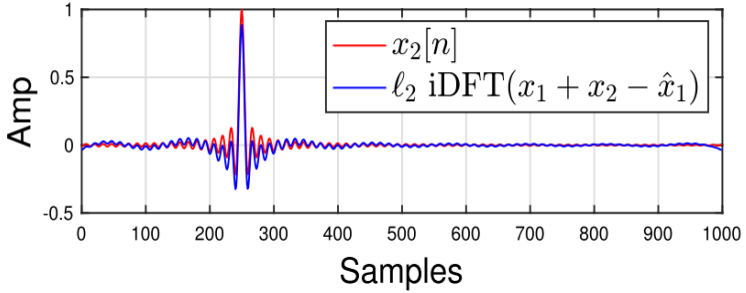
Fourier-series expansion-based signal representation

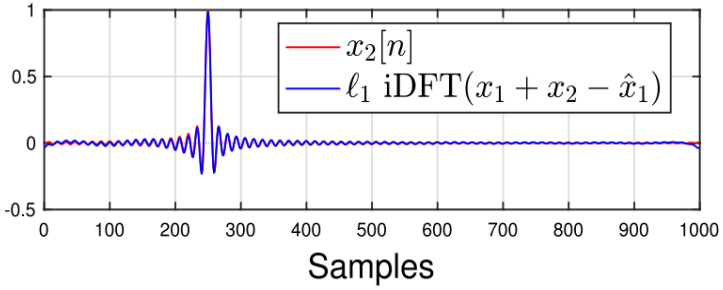
$x_2[n] = x_1[15(n+2.5)]$

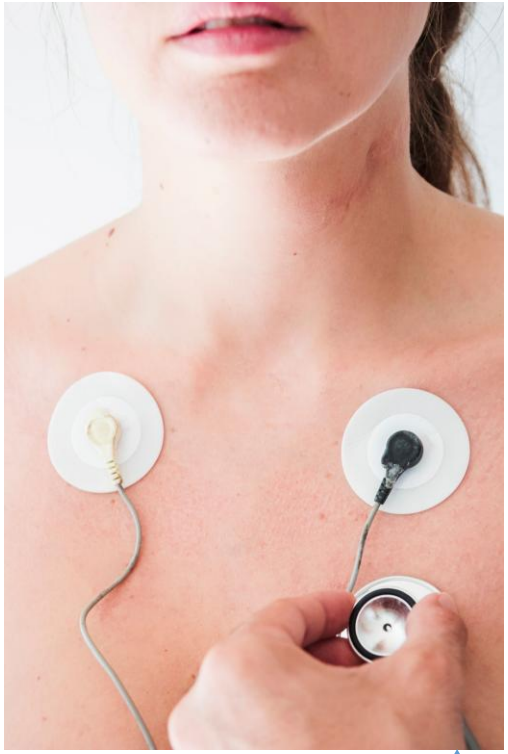




- x^1[n]is an estimate of x1[n]using the truncated Fourier-series either by ℓ2 or ℓ1 Fourier transform.







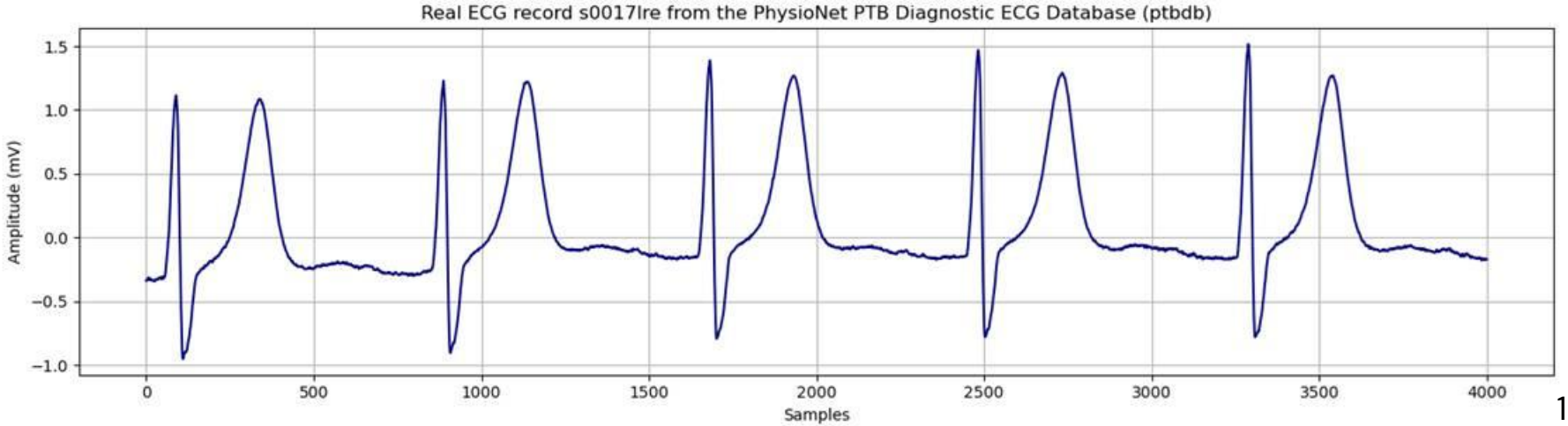
03

ECG Example

[redacted]

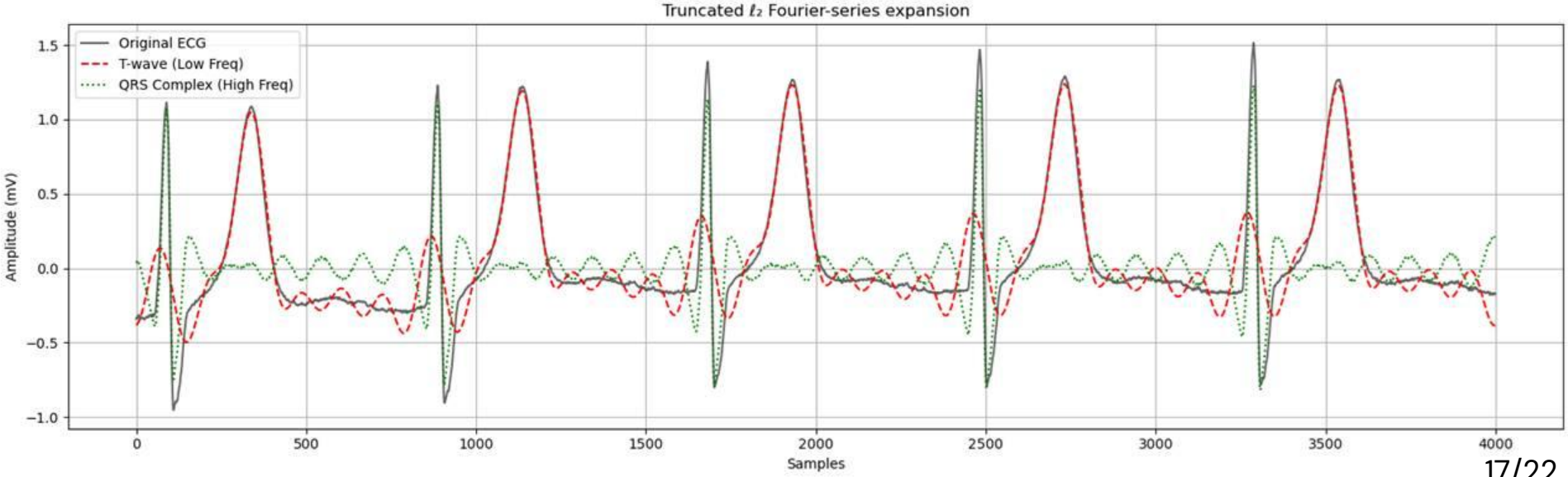
ECG Signal and Preprocessing

- Record: s0017lre,
- Sampling rate: 1000 Hz,
- Lead V3 (channel 8)
- Length: 4000
- samples = 4 seconds
- Bandpass filtering (0.5–40 Hz)



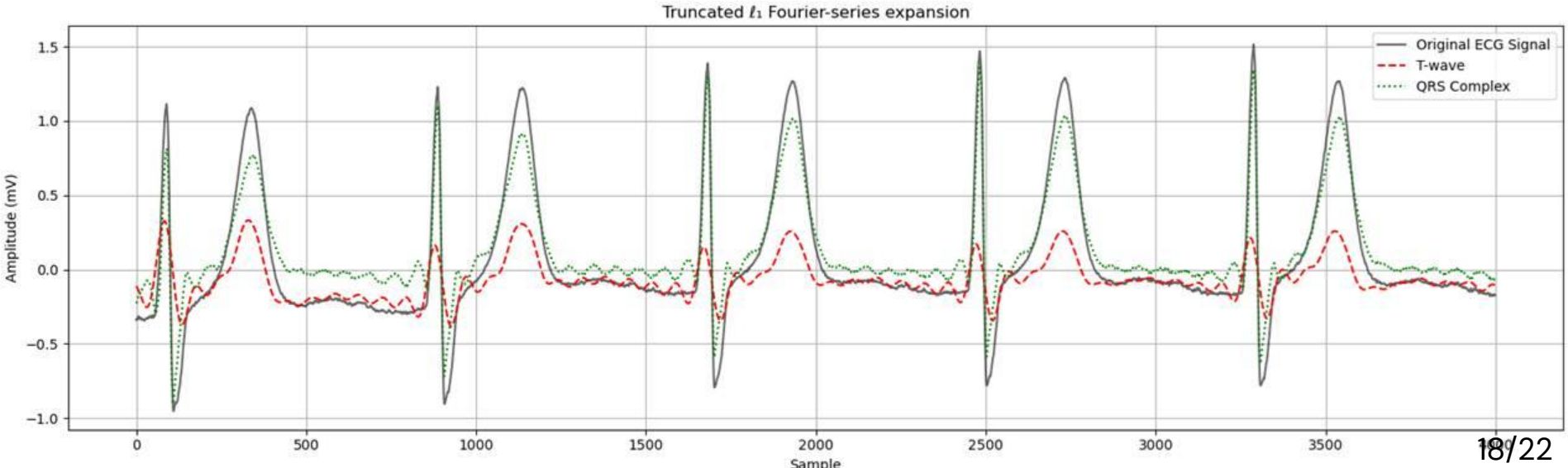
Truncated ℓ_2 Fourier-Series Expansion (M=32)

- Acts as a low-pass filter (cutoff ≈ 8 Hz)
- T-wave partially extracted
- Bandpass filtering (0.5–40 Hz)
- QRS components not rejected properly
- Gibbs phenomenon is observed (ringing near sharp edges)



Truncated ℓ_1 Fourier-Series Expansion (M=32)

- Improved separation of slow and fast components
- QRS complex suppressed
- T-wave extracted cleanly
- Less sensitivity to sharp peaks (impulses)
- Iterative optimization (MM algorithm)

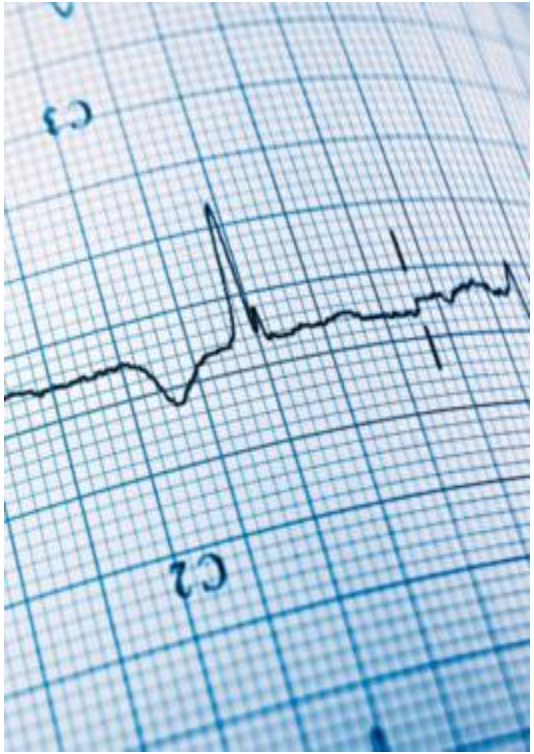


Required Python Packages

- ✓ To reproduce the simulation results and figures presented in this project, the following Python packages and versions were used

- wfdb==4.3.0
- numpy==2.2.5
- matplotlib==3.10.1

- ✓ pip install -r requirements.txt



04

Conclusion

[redacted]

ℓ_1 vs ℓ_2 Fourier Decomposition on ECG

Feature	ℓ_2 Fourier Series	ℓ_1 Fourier Series
Error minimization type	Minimizes **squared error** (Mean Squared Error)	Minimizes **absolute error** (Mean Absolute Error)
Assumed noise distribution	Gaussian (normal)	Laplace (impulsive / sparse)
Sensitivity to outliers	High – large errors dominate the result	Low – robust to outliers and spikes
Effect on Gibbs phenomenon	Strong – overshoots near discontinuities	Reduced – better handling of sharp transitions

ℓ_1 vs ℓ_2 Fourier Decomposition on ECG

Feature	ℓ_2 Fourier Series	ℓ_1 Fourier Series
Sparsity in coefficients	Smooth decay, no zero coefficients	Promotes sparsity – some coefficients may become zero
Performance for slow components	May mix slow and fast components (e.g., T & QRS overlap)	Better separation – preserves T-wave more accurately
Computation	Fast, explicit formula (FFT)	Requires iterative optimization (e.g., MM algorithm)
Interpretability in biomedical use	Less robust for ECG decomposition	Stronger at separating QRS and T-waves
Use cases	General signal processing	Signals with impulsive noise or overlapping frequency bands